RESEARCH PAPER

CUSTOMER ENGAGEMENT & RETENTION ANALYTICS FOR BANKING

SUMMARY

This project focuses on analyzing customer engagement, product utilization, and customer retention in the banking sector. The project uses customer data to understand customer behavior, identify churn patterns, and highlight customers who may require additional attention. Microsoft Power BI is used to transform raw data into interactive dashboards, KPI cards, charts, and tables. The analysis helps understand customer activity, retention rates, product usage, and customer risk indicators. The final dashboard supports data-driven decision-making and helps banks develop suitable customer retention strategies.

BUSINESS PROBLEM

Customer retention is an important challenge for banks because losing customers can affect revenue, customer relationships, and long-term business growth. Banks need to understand why customers leave, how customer activity relates to churn, and whether product usage is associated with retention.

Without proper analysis, it can be difficult to identify customer groups that may require targeted engagement. Therefore, this project analyzes customer data to identify churn patterns, evaluate product utilization, and support better retention planning.

OBJECTIVES

1. Analyze overall customer retention and churn rates.
2. Understand the relationship between customer activity and churn.
3. Analyze customer distribution according to the number of banking products used.
4. Compare retention patterns across different geographical regions.
5. Examine retention patterns across different age groups.
6. Identify inactive, high-balance customers who may benefit from additional engagement.
7. Develop an interactive Power BI dashboard for business reporting.
8. Provide recommendations to support customer retention strategies.

SCOPE OF THE PROJECT

The project focuses on customer engagement, churn, retention, and product utilization using banking customer data. It examines customer demographics, account balances, product counts, credit card ownership, activity status, and churn status.

The dashboard provides an overview of customer retention, product utilization, customer engagement, at-risk customer indicators, and retention recommendations. The findings can help banking teams understand customer behavior and plan targeted engagement activities.

DATASET DESCRIPTION

The project uses a European banking customer dataset containing approximately 10,000 customer records.

The dataset includes the following important fields:

- Customer ID: Unique identifier for each customer.
- Geography: Customer's geographical region.
- Age: Age of the customer.
- Balance: Customer's account balance.
- Number of Products: Number of banking products used by the customer.
- Has Credit Card: Indicates whether the customer owns a credit card.
- Is Active Member: Indicates whether the customer is an active banking customer.
- Exited: Indicates whether the customer has left the bank.
- Year: Year associated with the dataset.

The Exited column is used to distinguish churned customers from retained customers. An Exited value of 1 indicates a churned customer, while 0 indicates a retained customer. Similarly, IsActiveMember distinguishes active customers from inactive customers.

TOOLS AND TECHNOLOGIES USED

- Microsoft Power BI Desktop
- Power Query for data cleaning and transformation
- DAX for measures and calculated columns
- Microsoft Excel or CSV format for dataset storage
- Data visualization and business intelligence techniques
-

METHODOLOGY

The project follows a structured data analysis process.

Step 1: Data Collection

The banking customer dataset was obtained in CSV format and imported into Power BI Desktop.

Step 2: Data Understanding

The dataset was examined to understand its columns, data types, customer attributes, product information, and churn indicators.

Step 3: Data Cleaning and Transformation

Power Query was used to inspect data types, check for missing values, and prepare the dataset for analysis.

Step 4: Data Modeling and Calculations

DAX measures and calculated columns were created to calculate total customers, churned customers, churn rate, retention rate, activity status, age groups, and other analytical indicators.

Step 5: Dashboard Development

Interactive charts, KPI cards, tables, filters, and navigation buttons were created in Power BI.

Step 6: Analysis and Interpretation

Customer churn, retention, product utilization, geography, age groups, and engagement patterns were examined using the dashboard.

Step 7: Recommendations

Business recommendations were developed based on the patterns displayed in the analysis.

KEY PERFORMANCE INDICATORS (KPIs)

The dashboard includes the following key performance indicators:

1. Total Customers: Total number of unique customers in the dataset.
2. Churned Customers: Number of customers who have left the bank.
3. Churn Rate: Percentage of customers who have left the bank.
4. Retention Rate: Percentage of customers who have remained with the bank.
5. Product Depth Index: Average number of banking products used per customer.
6. Credit Card Retention Rate: Retention rate among customers who own a credit card.
7. High-Balance Disengagement Rate: Percentage of high-balance customers who are inactive.

These KPIs provide a quick overview of customer retention, engagement, and product utilization.

DASHBOARD DEVELOPMENT

The Power BI dashboard is divided into four pages to provide a clear and organized view of the analysis.

Page 1: Customer Retention Overview

This page presents the overall customer retention situation. It includes KPI cards for total customers, churned customers, and churn rate. Charts compare churn among active and inactive customers, show the distribution of retained and churned customers, and display churned customers by geography.

Page 2: Product Utilization Analysis

This page examines how the number of banking products relates to customer behavior. It includes churn rate by number of products, customer distribution by product count, and product portfolio distribution. These visuals help compare product usage patterns with customer retention.

Page 3: Customer Engagement & At-Risk Analysis

This page focuses on customer activity and customers identified by a rule-based at-risk indicator. It includes customer engagement status, at-risk customer counts, and a detailed table containing customer information such as customer ID, geography, age, balance, and number of products. This page helps identify customer groups that may benefit from proactive engagement.

Page 4: Retention Strength & Recommendations

This page compares retention rates across geographical regions and age groups. It also displays additional KPIs related to product depth, credit card retention, and high-balance customer inactivity. The page concludes with practical recommendations for improving customer engagement and retention.

DATA ANALYSIS AND FINDINGS

The dashboard supports analysis of several important customer behavior patterns.

Customer Retention: The churn rate and retention rate provide an overview of the proportion of customers who have left or remained with the bank.

Customer Activity: Comparing active and inactive customers helps examine how engagement status relates to churn.

Product Utilization: Analyzing the number of products used by customers helps reveal whether retention patterns differ between product-count groups.

Geographical Analysis: Comparing regions helps identify differences in customer retention across geographical groups.

Age-Based Analysis: Comparing age groups helps highlight customer segments with different retention patterns.

At-Risk Customer Analysis: A rule-based indicator identifies retained customers who are inactive and have balances at or above the dataset median. These customers can be considered for further review and targeted engagement.

The actual numerical findings should be reported using the values displayed in the completed Power BI dashboard. The at-risk indicator is a screening rule and should not be interpreted as a validated churn prediction model.

BUSINESS RECOMMENDATIONS

1. Develop targeted retention campaigns for customer groups with lower retention rates.
2. Contact inactive customers to understand their needs and encourage suitable banking engagement.
3. Offer relevant products based on customer requirements rather than promoting unnecessary products.
4. Review retention patterns across geographical regions and age groups.
5. Consider personalized services for inactive customers with relatively high account balances.
6. Evaluate credit card customer retention and improve relevant customer benefits.
7. Monitor churn and retention KPIs regularly to identify changes in customer behavior.
8. Use customer feedback and additional data to investigate the underlying reasons for churn.

BENEFITS OF THE PROJECT

- Provides a clear overview of customer retention and churn.
- Helps compare customer engagement across different groups.
- Makes product utilization patterns easier to understand.
- Supports the identification of customers who may need additional attention.
- Reduces manual reporting through interactive dashboards.
- Supports data-driven business decisions.
- Improves the presentation of customer analytics through visual reporting.

LIMITATIONS

The analysis depends on the available dataset and the information recorded in its columns. The dataset may not explain the actual reasons why customers leave the bank. The analysis identifies associations and patterns but does not establish that one factor causes churn.

The at-risk customer indicator is based on predefined rules rather than a machine learning prediction model. In addition, the dataset's year field is 2025, so the analysis should not be treated as a current view of the bank's customer base.

FUTURE SCOPE

Future improvements may include adding more recent customer data, transaction history, customer feedback, service interactions, and reasons for account closure. Machine learning techniques could be explored to develop and validate a churn prediction model. The project could also be extended with

automated reporting, more detailed customer segmentation, and a Streamlit application for sharing analytical insights.

CONCLUSION

The Customer Engagement & Retention Analytics project demonstrates how Power BI can be used to analyze banking customer behavior and present useful business insights. The dashboard combines customer retention, churn, product utilization, geography, age groups, and engagement indicators in one interactive reporting solution.

The analysis helps highlight customer segments that may require further investigation and supports the development of targeted retention strategies. Although the dashboard does not independently predict future churn, it provides a foundation for monitoring customer behavior and making data-informed decisions. Overall, the project demonstrates the practical application of data cleaning, DAX calculations, KPI development, and dashboard visualization in business analytics.

REFERENCES

1.Company-provided financial european banking dataset used for the analysis.

2.Microsoft Power BI Documentation – Microsoft – Power BI features, data analysis, and visualization.

3.Microsoft Power Query Documentation – Microsoft – Data cleaning and transformation